Week 12 Milestone Report

Graph Analytics and Centrality Report

MovieLens Discovery System — Semester Project

Grupo 06 — Big Data

`BD-Grupo-06/movielens-discovery`

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Contents

1 Executive Summary

This report covers Week 12: formalizing the item-item graph induced by the MovieLens catalog and using it for structural analysis. The graph reuses the Week 10 SVD item-factor space (the same signal behind the `svd_collaborative` recommender), so this is a genuine structural view of an existing signal, not an unrelated ad hoc graph.

Table 1: Graph overview and headline validity/comparison numbers

Metric	Value
Nodes	13,176 (movies with $\geq$ 50 ratings, i.e. a W
Edges	155,331 (undirected, weighted by cosine si
Sparsity	0.179% of all possible pairs
Isolated nodes	29 (0.22%)
Connected components	33
Giant component coverage	13,139 nodes (**99.72%**)
Spearman ρ: popularity vs. graph PageRank	**0.313**
Spearman ρ: model rec. in-degree vs. graph PageRank	**0.578**

The graph is highly connected (one dominant component covering 99.7% of nodes) at the chosen configuration (k = 15 neighbors, minimum cosine similarity 0.5). PageRank correlates only weakly with raw popularity ($\rho = 0.313$), confirming that graph centrality captures something structurally different from "most-rated" — a small number of niche, low-popularity titles are highly central because their rating pattern places them in a densely interconnected similarity neighborhood.

2 Objective

The goal of Week 12 was to formalize the domain as a graph and use it for structural analysis, connecting back to the Week 10 recommendation layer rather than building an unrelated graph:

> *What does the similarity structure among movies look like as a graph, and does structural centrality tell us something different from popularity or model-based ranking?*

Specific questions addressed:

- What do nodes and edges mean, and why are they defined this way?
- Is the chosen graph configuration (k, similarity threshold) robust, or a hand-picked lucky run?
- Does PageRank on the similarity graph rank movies differently than popularity or the Week 10 recommender's in-degree?
- What does high centrality mean in this domain, and — just as important — what does it *not* mean?

3 Inputs and Reproducible Artifacts

3.1 Input artifacts from previous weeks

Table 2: Inputs consumed from prior weeks

Artifact	Source	Purpose
`week10_svd_item_factors.parquet`	Week 10	50-dim item factors: node features / edge weig
`week10_popularity_global.csv`	Week 10	Popularity ranking (comparison baseline)
`week10_svd_recs_top20.parquet`	Week 10	Recommendation in-degree (model-based basel
`week07_kmeans_assignments.csv`	Week 7	Cluster labels for the web viewer
`movies_catalog.parquet`, `links.csv`	Week 3	Titles, genres, IMDb/TMDb IDs

3.2 Week 12 output artifacts

Table 3: Output artifacts produced in Week 12

File	Description
`week12_graph_meta.json`	Graph definition, chosen configuration, validity cl
`week12_graph_metrics.parquet`	Per-node degree, weighted degree, PageRank, eig
`week12_sensitivity_sweep.csv/.html/.png`	Sweep over $k \in \{5, 10, 15, 20\} \times$ threshold $\in \{0.3,$
`week12_popularity_vs_pagerank.html/.png`	Popularity rank vs. graph PageRank rank scatter
`movie_graph_viz.json`	Sampled 400-node / 3,334-edge subgraph for the

4 Graph Definition

Node: a movie with at least 50 ratings in the 25M-rating matrix, i.e. a movie that has a Week 10 SVD item factor (`min_movie_ratings=50` per `week10_svd_meta.json`). Restricting nodes to this set — rather than all ≈62,423 catalog rows — is a deliberate grain choice: movies below the rating floor have no reliable collaborative signal, so a "similarity" edge to them would be noise, not structure. This yields **13,176 candidate nodes**.

Edge: undirected, weighted. Weight = cosine similarity between two movies' 50-dimensional SVD item-factor vectors — the same vectors and metric used by the Week 10 `svd_collaborative` recommender. An edge exists between A and B if B is among A's top-k nearest neighbors by cosine similarity **or** A is among B's top-k neighbors (symmetrized), and the similarity clears a minimum threshold.

Why undirected. Cosine similarity is symmetric by construction ($\text{sim}(A, B) = \text{sim}(B, A)$), so a directed graph would only be justified by an asymmetric relation. That asymmetric relation ("B appears in A's top-20 recs but not vice versa") is kept as a *separate* ranking signal (`model_rec_indegree`) for the comparison in Section 8, rather than being conflated with the edge definition.

Why weighted and thresholded, not fully dense. A fully dense weighted graph on 13,176 nodes (≈86.8M possible undirected pairs) would be neither interpretable nor visualizable, and most pairs have near-zero similarity. Keeping only the top-k neighbors per node above a minimum similarity controls sparsity and keeps only edges that plausibly reflect a real co-rating relationship. Both k and the threshold are swept in Section 5 rather than chosen by hand.

5 Graph Construction and Sensitivity Sweep

Edges are built with `sklearn.neighbors.NearestNeighbors` (`metric='cosine'`, brute force), which computes each node's k nearest neighbors without materializing the full 13,176 × 13,176 similarity matrix, then symmetrizes and drops pairs below the minimum similarity.

k and the minimum-similarity threshold were swept jointly to check whether the graph's structure is sensitive to the choice, rather than reporting one hand-picked, unexamined configuration (Table 4).

Table 4: Sensitivity sweep: edges, sparsity, isolated nodes, and connectivity by k and threshold. Chosen configuration in bold.

k	min. sim.	Edges	Sparsity	Isolated	Components	Giant frac.
5	0.3	54,494	0.063%	1	2	99.99%
5	0.5	54,036	0.062%	29	33	99.72%
5	0.7	41,172	0.047%	1,903	1,957	84.56%
10	0.3	107,115	0.123%	1	2	99.99%
10	0.5	105,447	0.121%	29	33	99.72%
10	0.7	74,813	0.086%	1,903	1,957	84.56%
15	**0.5**	**155,331**	**0.179%**	**29**	**33**	**99.72%**
15	0.3	158,911	0.183%	1	2	99.99%
15	0.7	104,760	0.121%	1,903	1,957	84.56%
20	0.3	210,185	0.242%	1	2	99.99%
20	0.5	203,980	0.235%	29	33	99.72%
20	0.7	132,124	0.152%	1,903	1,957	84.56%

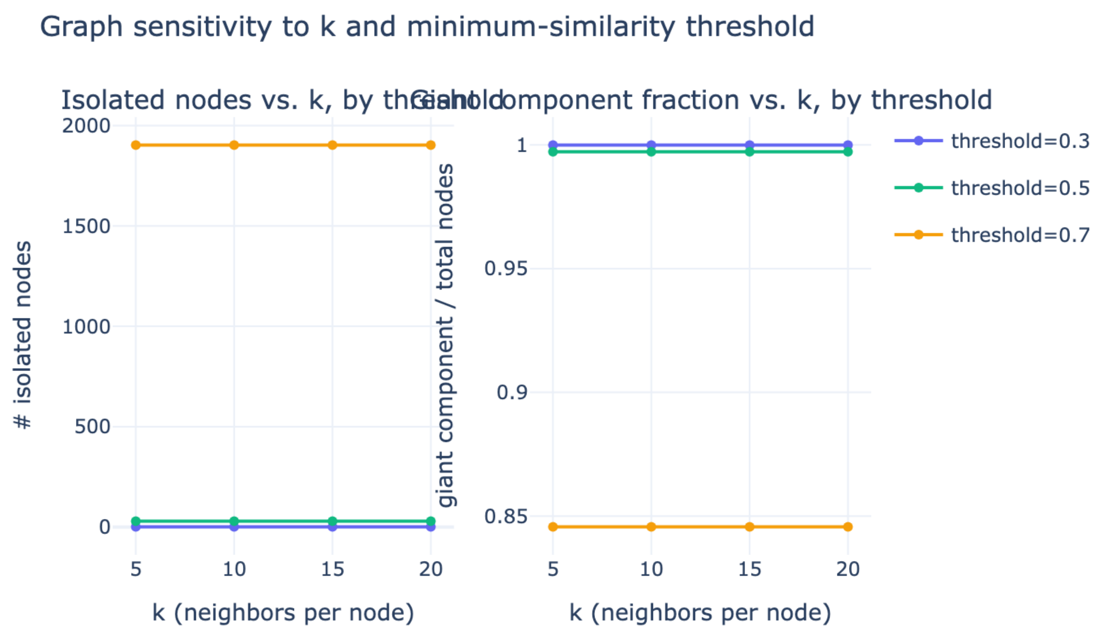

Figure 1: Isolated-node count and giant-component fraction as functions of k, grouped by minimum-similarity threshold.

Key observation. For a fixed threshold, the isolated-node count is *constant* across all values of k (1 at threshold$=0.3$, 29 at threshold$=0.5$, 1,903 at threshold$=0.7$). This is because isolation is driven by whether a node's single closest neighbor clears the threshold at all — if the top-1 neighbor's similarity already falls below the threshold, looking at more neighbors (k) cannot produce a qualifying edge. The threshold, not k, controls connectivity; k mainly controls how many *additional* edges accumulate once a node is already connected.

6 Chosen Configuration and Validity Checks

Threshold$=0.7$ strands 1,903 nodes (14.4%) even at $k = 20$ — too strict, since most real co-rating similarity signal in this 50-dimensional space falls below 0.7. Threshold$=0.3$ is already near-fully-connected at $k = 10$ (99.99% giant component, only 1 isolated node) — too permissive, approaching a near-complete graph and defeating the purpose of a sparse structural graph.

Chosen configuration: $k = 15$, min_similarity$= 0.5$. This keeps the giant component covering the large majority of nodes (99.72%) while still discarding most low-similarity pairs (sparsity 0.179%), and matches how the Week 10 `content_cosine` / `svd_collaborative` recommenders already treat similarity — same latent space, same metric.

Table 5: Validity checks at the chosen configuration

Check	Result
Edge sparsity	0.179% of all possible undirected pairs — sparse by design
Isolated nodes	29 / 13,176 (0.22%)
Connected components	33 (one giant component + 32 small fragments)
Giant component coverage	13,139 / 13,176 nodes (99.72%)
Sensitivity	Section 5 — connectivity is threshold-driven, not one lucky k

7 Centrality Measures

Degree, weighted degree, and PageRank are computed on the full graph. Eigenvector centrality is undefined on disconnected graphs (`networkx` raises `AmbiguousSolution`), so it is computed on the giant connected component only, with 0 assigned to the 37 nodes outside it — those nodes have no path into the component's dominant eigenvector by construction, so 0 is the correct value, not a missing one. Exact betweenness centrality is $O(V \cdot E)$ and infeasible at this scale (13,176 nodes, 155,331 edges); an approximate betweenness is computed instead using a random sample of 500 source nodes (`networkx`'s native `k` parameter), with edge distance defined as $1 - \text{similarity}$.

Table 6: Top 10 movies by PageRank, with degree, weighted degree, and popularity/model rank (out of 13,176)

Movie	PageRank	Degree	Wtd. degree	Pop. rank	Model-rec
L'Atalante (1934)	0.000265	107	100.32	6,351	
The Lazarus Effect (2015)	0.000261	96	85.01	7,985	
Viridiana (1961)	0.000247	101	95.56	5,592	
Day of Wrath (1943)	0.000237	94	88.33	8,052	
Perfect Stranger (2007)	0.000230	82	68.94	6,044	
Untraceable (2008)	0.000228	81	66.64	5,594	
The Reaping (2007)	0.000226	82	70.13	5,950	
Pickpocket (1959)	0.000223	90	84.13	6,120	
I Know Who Killed Me (2007)	0.000223	79	70.20	7,632	
Devil's Knot (2013)	0.000217	77	61.85	10,766	

Key finding. None of the top-10 PageRank movies rank in the popularity top-1,000 — the highest is L'Atalante at popularity rank 6,351 out of 13,176 (bottom half of the catalog). High PageRank here means a movie sits in a densely-interconnected neighborhood of the similarity structure (reachable through many strong-similarity chains), which is a structurally different property from being widely rated.

8 Comparison: Graph Ranking vs. Popularity vs. Model-Based Ranking

Three independent rankings over the same 13,176-node set:

- **Popularity ranking**: rank by `rating_count` (Week 10 `week10_popularity_global.csv`) — "how many people rated this."
- **Model-based ranking**: rank by *recommendation in-degree* — how many times a movie appears inside another movie's Week 10 SVD top-20 recommendation list. This is a directed, model-derived signal distinct from the graph's undirected similarity edges.
- **Graph ranking**: rank by PageRank on the Week 12 similarity graph.

Table 7: Spearman rank correlations between the three ranking signals

Comparison	Spearman ρ
Popularity vs. graph PageRank	**0.313**
Model rec. in-degree vs. graph PageRank	**0.578**
Popularity vs. model rec. in-degree	0.785

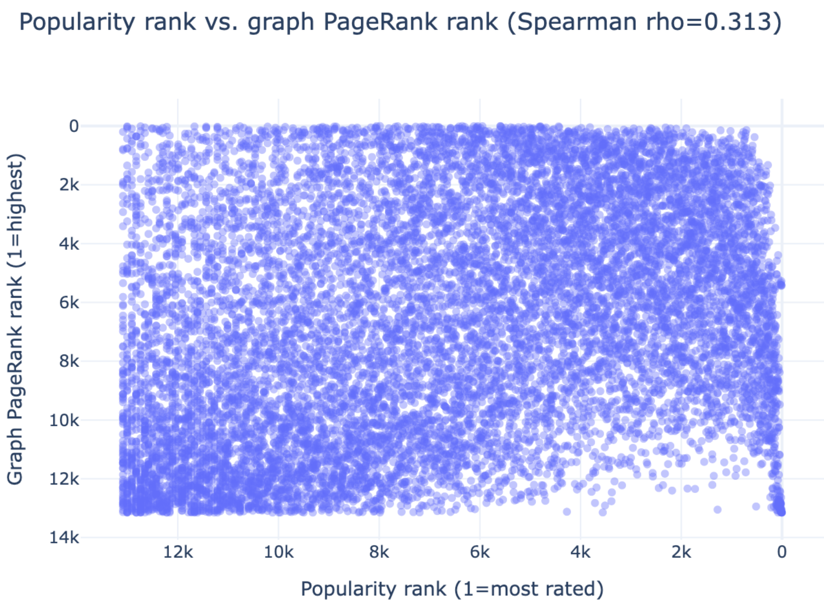

Figure 2: Popularity rank vs. graph PageRank rank (Spearman $\rho = 0.313$). Points scatter widely around the diagonal, showing the two rankings diverge substantially.

Popularity and the model's recommendation in-degree are strongly correlated ($\rho = 0.785$) — both are, in different ways, proxies for how mainstream a movie is. Graph PageRank correlates much more weakly with popularity ($\rho = 0.313$) and moderately with model in-degree ($\rho = 0.578$): the similarity graph surfaces a partially different signal — structural embeddedness in the SVD similarity space — that is only loosely tied to how many people rated a movie.

9 Interpretation Note: What Graph Structure Means, and Does Not Mean

What it means: an edge between two movies indicates that, in the space learned by the Week 10 SVD factorization of the user-rating matrix, the two movies have highly correlated rating patterns across the user base — people who rate one in a particular direction tend to rate the other similarly. High PageRank means a movie sits in a densely-interconnected neighborhood of this similarity structure: not just similar to a few movies, but reachable through many short chains of strong similarity.

What it does not mean:

- It is **not** a claim about shared plot, cast, or genre — the graph is built purely from rating co-occurrence, not content features. Two movies can be structurally central neighbors while sharing no genre tag.
- High PageRank is **not** the same as "popular" — $\rho = 0.313$ is positive but far from 1: the graph surfaces movies that are structurally embedded in the similarity space without necessarily being the most-rated movies overall, and vice versa.
- The graph is **not directed** and does not encode "recommend A because of B" — that asymmetric signal is captured separately by the Week 10 recommendation lists (`model_rec_indegree`) used for comparison above.
- The similarity signal inherits whatever rating biases exist in MovieLens (predominantly English-language, enthusiast-skewed userbase); it should not be read as a universal notion of movie similarity.
- Isolated nodes and small disconnected components are not "unrelated" movies in general — they are movies whose rating pattern did not clear the chosen similarity threshold against any node's top-15 neighbors, which can also reflect a small or unusual rater population for that title.

10 Subgraph Sampling for the 3D Web Visualization

Drawing all 13,176 nodes in a browser-based 3D scene is neither readable nor performant, so a **connected** subgraph of 400 nodes was sampled for the interactive viewer (`web/`) using hub-expansion sampling: start from the 40 highest-PageRank nodes in the giant component (structural hubs), then greedily grow the sample by always adding the unselected node with the strongest edge weight to an already-selected node. Because every added node is attached by construction, the sampled subgraph has **no isolated nodes**, unlike a uniform-random node sample with an induced-edge cut, which typically strands many isolated points.

Table 8: Full graph vs. sampled subgraph used by the web viewer

	Full graph	Sampled subgraph
Nodes	13,176	400
Edges	155,331	3,334
Isolated nodes	29	0 (by construction)

The sampled subgraph is exported to `artifacts/week12/movie_graph_viz.json` and copied to `web/public/movie_graph.json`, where node metadata (title, genres, IMDb/TMDb links, PageRank, degree, cluster ID) drives the interactive viewer. Poster URLs are filled in separately by `scripts/fetch_posters.py` via the TMDb API, for exactly this sampled set of movies.

11 What Worked and What Did Not

What worked.

- Reusing the Week 10 SVD item-factor space as the graph's edge weight keeps the graph analysis connected to an existing, already-validated signal instead of introducing an unrelated ad hoc similarity metric.
- The sensitivity sweep cleanly separates the role of k (marginal edge accumulation) from the role of the similarity threshold (connectivity), which made the configuration choice defensible rather than arbitrary.
- Hub-expansion sampling produced a fully connected, hub-anchored 400-node subgraph suitable for a real-time 3D viewer, with zero isolated nodes.

What did not work.

- **Eigenvector centrality** is undefined outside the giant component; 37 nodes necessarily receive a 0 that reflects "no path into the dominant eigenvector," not "zero centrality" in an absolute sense — this distinction has to be stated explicitly or it reads as a missing value.
- **Betweenness centrality** could not be computed exactly at this scale (13,176 nodes, 155,331 edges); the 500-source approximation trades precision for tractability and should not be read as an exact ranking.
- **Weak correlation with popularity** ($\rho = 0.313$) means graph PageRank alone is a poor stand-in for "movies most people already know" — it answers a different question (structural embeddedness), not "what should the homepage show a new user."

12 Ethics and Access Note

- **Data source**: GroupLens MovieLens 25M public release (research license).
- **Access**: Educational and research use; raw data not redistributed.
- **Personal data risk**: all user identifiers are anonymized integer IDs.

- **Mitigation**: the graph is built entirely at the movie level from aggregate SVD item factors; no individual user rating history or user identifier is exposed by any node, edge, or centrality score in this analysis.

13 Reproducibility

Run the notebook after the Week 10 artifacts exist:

1. `notebooks/week12/week12_graph_analytics.ipynb`

Then, to refresh the web viewer's data:

2. `python scripts/fetch_posters.py` — populates `posterUrl` for the ≈400 sampled movies via the TMDb API.
3. Copy `artifacts/week12/movie_graph_viz.json` to `web/public/movie_graph.json`.

Table 9: Key parameters for reproducibility

Parameter	**Value**
Node definition	Movie with ≥ 50 ratings (has a Week 10 SVD item factor)
Candidate nodes	13,176
Similarity metric	Cosine, on 50-dim Week 10 SVD item factors
Chosen k (neighbors per node)	15
Chosen minimum similarity	0.5
Betweenness sample size	500
Subgraph sample size (web viewer)	400 nodes / 3,334 edges
Hub seeds for sampling	40 (highest PageRank in giant component)
Random state	42

14 Conclusion

Week 12 is complete. A 13,176-node, 155,331-edge item-item similarity graph was built directly from the Week 10 SVD item-factor space, validated with a joint sensitivity sweep over k and similarity threshold, and analyzed with degree, weighted degree, PageRank, eigenvector, and approximate betweenness centrality.

- The chosen configuration ($k = 15$, min_similarity$= 0.5$) yields a giant component covering 99.72% of nodes, and the sweep shows this connectivity is driven by the threshold, not a lucky choice of k.
- Graph PageRank correlates only weakly with popularity ($\rho = 0.313$) and moderately with the Week 10 model's recommendation in-degree ($\rho = 0.578$): structural centrality in the similarity graph is a genuinely different signal from "most-rated" or "most-recommended," not a relabeling of either.
- The interpretation note makes explicit what the graph does and does not license: it reflects rating co-occurrence, not content similarity; it is undirected and does not encode asymmetric recommendation; and it inherits MovieLens's own rating-population biases.

- A connected, hub-anchored 400-node subgraph was sampled and exported to drive the interactive 3D MovieLens Discovery viewer (`web/`), completing the pipeline from raw ratings through representation, clustering, recommendation, and now graph structure.